

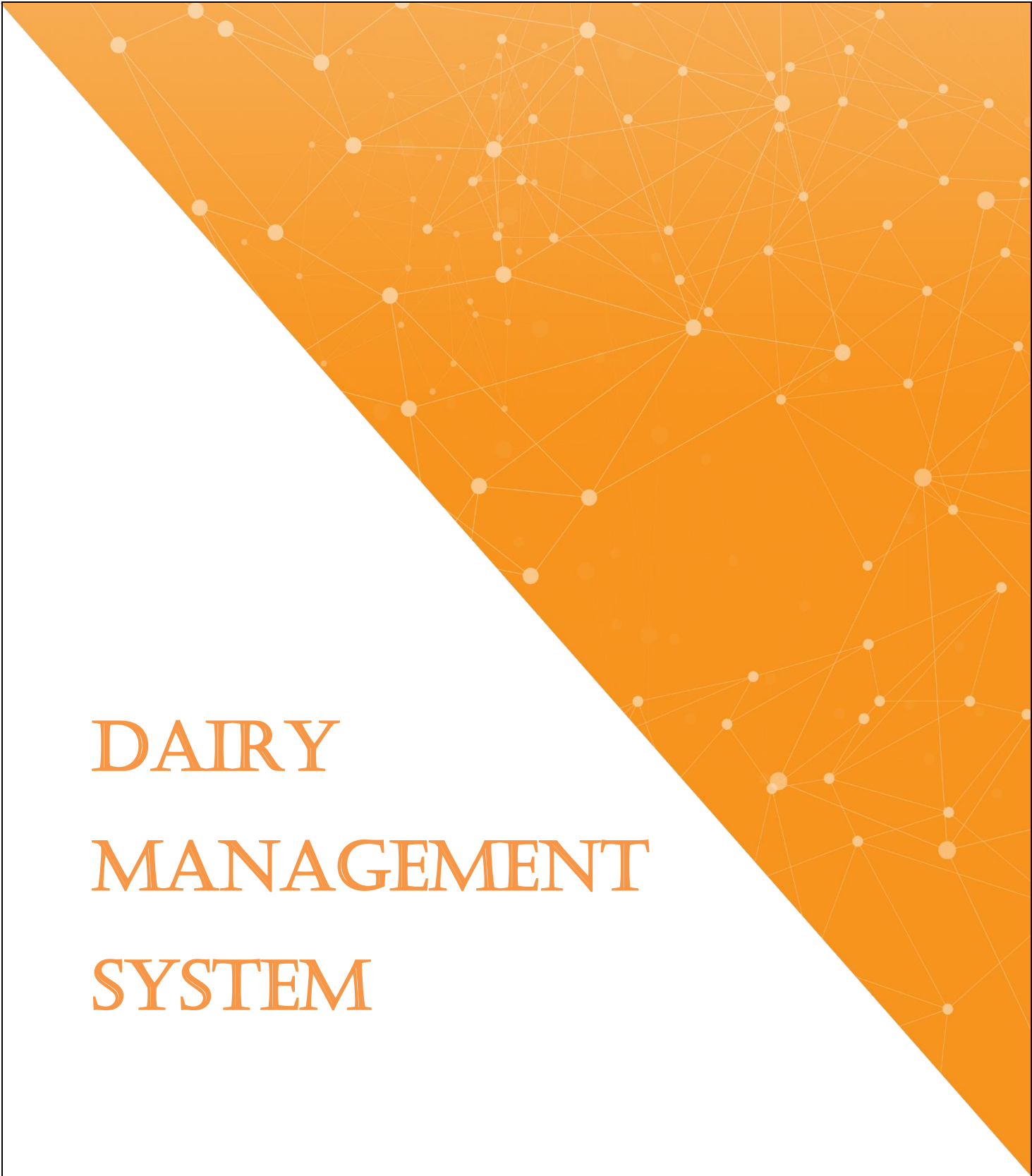
JADAVPUR UNIVERSITY

MCA - 1ST YEAR 2ND SEMESTER 2024

LAB ASSIGNMENT - DATABASE MANAGEMENT SYSTEM

NAME – Shruti Mondal

ROLL - 002310503030



DAIRY MANAGEMENT SYSTEM

Dairy Management System

By

Shruti Mondal (002310503030)

Suneha Dutta (002310503048)

ABSTRACT

An old method of keeping records of dairy members using pen and paper is not reliable as finding old data can take a long time. The Dairy Management System provides dairy owners an easy way to organize the data of dairy members in one place through a web application. Through the dairy management system, the dairy owner can keep records of the individual's daily milk collection and the same will be credited to the user's account so that both the owner and the user can keep track of past records. Also, the dairy owner gets a few more options in the dashboard. Only the owner can verify the user's account after the user registers in the system. With this, the owner can keep track of daily total collection, total revenue, total expenses, collection between two dates. Both the user and the owner can receive receipts from their account for milk related contributions and collection related transactions respectively.

PROBLEM STATEMENT

Most dairy owners keep their records in register books which is not reliable as it may be lost somewhere or a little difficult to manage. Also, finding records of each individual's specific period is a time-consuming process. Using a dairy management system, the dairy owner can keep a record of all dairy members by organizing data in one place and this is the most reliable system as the owner can access the dashboard from any device.

INTRODUCTION

The Dairy Management System (DMS) is a web based application that can be accessed over the web. This system can be used to automate the workflow of dairy shop and their invoices.

The project has been planned to be having the view of distributed architecture, with centralized storage of the database. The application for the storage of the data has been planned. Using the constructs of MySQL Server and all the user interfaces has been designed using the PHP technologies. The database connectivity is planned using the "MySQL Connection" methodology.

DMS is a web-based application which manages the products of dairy shop. It has one module i.e. admin who manages all the functions of the dairy shop.

OBJECTIVE

- To Provide Basic Facility Manage Organization Schedule and work.
- This project is aimed at developing a web based Dairy Management System Tool, which is of importance to either a small dairy shop or big dairy shop. The Dairy Shop Management System is a software based application works as a simple Dairy Software to maintain daily milk record and maintain reports.

MANUAL VS PROPOSED SYSTEM

Manual System

When we Analysis the Manage about this firm then we face that they working with manual. And we all know that the Manual System has many disadvantages. Some are mentioned below...

- The manual system requires more time for processing.
- It requires more critical work.
- The manual system is more error prone.
- Difficult to maintain.
- Manual system is costly.
- Immediate response to the queries is difficult and time consuming.
- More men power needed.
- Manual system show of the particular place.

Proposed System

Proposed system is required because of some advantages of new system are as below...

- The Proposed system required less time for completion of any work.
- Proposed system should work smoothly and very fast.
- Proposed system saving time and manpower.
- The System is user friendly and anyone having computer knowledge
- Can handle it easily.
- Suitability for computerized data entry.
- New system is decreasing the chances of error.
- Maintaining Dairy information,
Staff information & Customer information, Milk Rate
Information.

DOMAIN DESCRIPTION

This project is based on an online dairy management system where farmers can register and add details about their cattle and put information on the dairy products they sell. Whereas, customers can buy the products from them and all the whole process is being monitored or managed by the employees who also work as the admin.

MODULES

1.Registration System:

Both the customers and the farmers can register themselves by providing the basic details.

2.Login System:

Users can login into the system, they can update passwords, manage their accounts. Farmers can manage the information about the products and customers can buy products or give feedback by logging into the system.

3.Buying of Dairy products:

The main module of this system is buying and selling off the dairy products, and rating them.

4.Transaction System:

Transaction system is also a module of this system, which eases this whole project.

Functional and Operational Requirements

Requirement analysis is performed by the senior members of the team with inputs from the customer, the sales department, market surveys and domain experts in the industry. Information regarding what kind of input is required and what output is expected is collected during this stage, and is then used to plan the basic project approach and to conduct product feasibility in the economical, operational and technical areas. Planning for the quality assurance requirements and identification of the risks associated with the project is also done.

The first step in the system development life cycle is the preliminary investigation to determine the feasibility of the system. The purpose of the preliminary investigation is to evaluate the project request. It is neither a design study nor does it include the collection of details to describe the system in all respect. Rather it is collection of information that helps the maker to evaluate the merits of the project requests and make an informed judgment about the proposed project. The report analyses accomplish the following objective:

- Clarifying and determining the project request.
- Determining the size of the project.
- Accessing costs and benefits of alternatives approaches.
- Reports of the finding to the organization, with recommendations outlining the acceptance of the proposal.

Reviewing Organization document: In this, the investigation is made about the organization involved in, or affected by the project. So, the present activities are studied in order to develop a fully automated system. For this, feasibility study is done.

Feasibility Study

Feasibility Study is an analysis that takes all of the project factors into account – including Operational, Technical, Economic and Time Feasibility studies to ascertain the likelihood of completing the project successfully.

Operational Feasibility

Proposed project is beneficial only if it can be turned into information systems that will meet the operating requirements of the organization. This feasibility asks if the system will work when it is developed and installed. So, in this project we tested the operational feasibility by determining:

- Whether there is sufficient support for the project from the organization.
- Is the current system being acceptable by them?
- Will the proposed system cause harm?
- Will customers be satisfied with the system?
- Will suppliers always be able to provide right information about the information about the vehicle all times?
- Will the service of the system be enough according to customers' demands?

Technical Feasibility

There are number of technical issues which are generally raised during the technical feasibility. They are determined as follows:

- Does the necessary technology assist to do what is suggested?
- Will the system be easily accessible?
- Can customers easily find the vehicles they want to book?
- How easy is the process of booking of a vehicle?
- Can the system be upgraded once developed?
- Security of the system.
- Regular updating of the system.

Economic Feasibility

A system that can be developed technically and that will be used if installed must still be profitable for the organization. The first purpose of this effort is to financially model the venture opportunity and achieve a break-even analysis. These scenarios will be critical in strategic planning, milestone development and venture valuation analysis. Financial benefits must equal or exceed the costs. The analysis raises various financial questions during the investigation to estimate the following:

- The cost of conduct, a full system investigation.
- The cost of hardware and software for the application being used.
- The benefits in the form of cost or fewer costly errors.
- The cost if nothing changes.

To be judged feasible, a proposal for the specific project must pass all these tests. Otherwise, it is not considered as economically feasible. In general, requests that do not pass all the feasibility tests are not persuaded further, unless they are modified and resubmitted as new proposals.

Time Feasibility

The required systems development was expected to be successfully completed within the stipulated time, as the constraints of time placed on the process was of crucial importance. Thus, it can be finally concluded that the proposed project fulfilled all feasibility requirements, the implementation of the project was progressed with. Thus, everything has been done while taking time into consideration.

Language Used

Frontend

- HTML (***HYPER TEXT MARKUP LANGUAGE***)
- CSS (***CASCADING STYLE SHEET***)
- BOOTSTRAP (***FRAMEWORK OF CSS AND JAVASCRIPT***)

Backend

- PHP (***HYPERTEXT PREPROCESSOR***) (***SERVICE SLIDE LANGUAGE***)
- MYSQL (***DATABASE***)

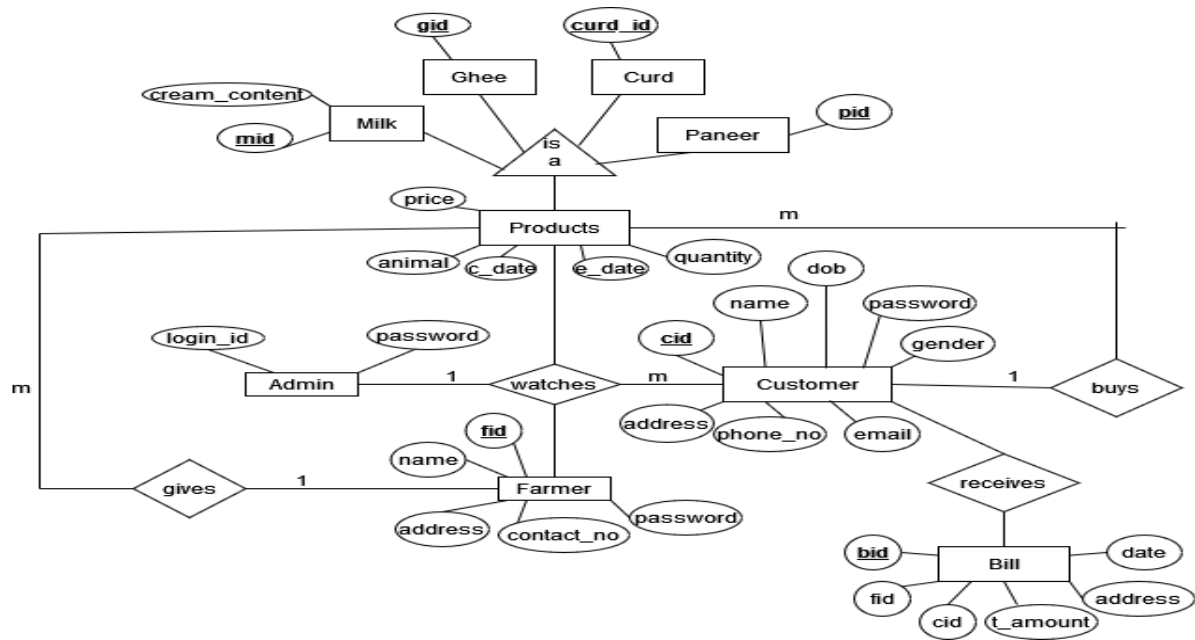
ER DIAGRAM

Entity and Attributes:

Entity	Attributes
Admin	<i>login_id</i> , password
Farmer	<i>fid</i> , name, address, contact_no_ password
Customer	<i>cid</i> , name, dob, gender, password, email, address, phone_no
Products	price, animal, c_date, e_date, quantity
Milk	<i>mid</i> , cream_content
Ghee	<i>gid</i>
Curd	<i>curd_id</i>
Paneer	<i>pid</i>
Bill	<i>bid</i> , fid, cid, t_amount, address, date

Relationships:

- Admin ***Watches*** Customer
- Admin ***Watches*** Farmer
- Customer ***Receives*** Bill
- Farmer ***Gives*** Products
- Customer ***Buys*** Products



ER DIAGRAM OF DAIRY MANAGEMENT SYSTEM

SAMPLE DATA FOR THESE ENTITIES:

1. ADMIN

login_id	password
admin1	pass1234
admin2	secure5678

2. FARMER

fid	name	address	contact_no	password
1	Rajesh Kumar	123 Village Road, Karnataka	9876543210	farmer123
2	Sunita Sharma	456 Farm Street, Maharashtra	9876543211	farm5678

3. CUSTOMER

cid	name	dob	gender	password	email	address	phone_no
1	Priya Singh	1990-01-15	Female	cust1234	priya@example.com	789 City Avenue, Delhi	9876543212
2	Amit Patel	1985-05-23	Male	patel5678	amit@example.com	321 Town Street, Gujarat	9876543213

4. PRODUCTS

price	animal	c_date	e_date	quantity
150	Cow	2024-05-01	2024-05-15	100
100	Goat	2024-05-03	2024-05-18	150

5. MILK

mid	fat_content
1	3.5%
2	4.0%

6. PANEER

pid
1
2

7. CURD

cid
1
2

8. GHEE

gid
1
2

9. BILL

bid	fid	cid	t_amount	address	date
1	1	1	200	789 City Avenue	2024-05-10
2	2	2	150	321 Town Street	2024-05-11

Assignment 1

Shruti Mondal

Master of Computer Application

Roll No. : 002310503030

Semester : II

1)

-- Creating the EMPLOYEE table

```
CREATE TABLE EMPLOYEE (  
    EMP_CODE CHAR(16) PRIMARY KEY,  
    EMP_NAME CHAR(20),  
    DEPT_CODE CHAR(16),  
    DESIG_CODE CHAR(16),  
    SEX CHAR(1),  
    ADDRESS CHAR(25),  
    CITY CHAR(20),  
    STATE CHAR(20),  
    PIN CHAR(6),  
    BASIC NUMBER,  
    JN_DT DATE  
);
```

-- Creating the DESIGNATION table

```
CREATE TABLE DESIGNATION (  
    DESIG_CODE CHAR(16) PRIMARY KEY,  
    DESIG_DESC CHAR(20)  
);
```

-- Creating the DEPARTMENT table

```
CREATE TABLE DEPARTMENT (  
    DEPT_CODE CHAR(16) PRIMARY KEY,  
    DEPT_NAME CHAR(20)  
);
```

2)

SQL> DESCRIBE EMPLOYEE;

Name	Null?	Type
EMP_CODE	NOT NULL	CHAR(16)
EMP_NAME		CHAR(20)
DEPT_CODE		CHAR(16)
DESIG_CODE		CHAR(16)
SEX		CHAR(1)

ADDRESS	CHAR(25)
CITY	CHAR(20)
STATE	CHAR(20)
PIN	CHAR(6)
BASIC	NUMBER
JN_DT	DATE

SQL> DESCRIBE DESIGNATION;

Name	Null?	Type

DESIG_CODE	NOT NULL	CHAR(16)
DESIG_DESC		CHAR(20)

SQL> DESCRIBE DEPARTMENT;

Name	Null?	Type

DEPT_CODE	NOT NULL	CHAR(16)
DEPT_NAME		CHAR(20)

3)

-- Inserting rows into the DESIGNATION table

```
INSERT INTO DESIGNATION (DESIG_CODE, DESIG_DESC) VALUES ('MGR', 'Manager');
INSERT INTO DESIGNATION (DESIG_CODE, DESIG_DESC) VALUES ('EXE', 'Executive');
INSERT INTO DESIGNATION (DESIG_CODE, DESIG_DESC) VALUES ('OFF', 'Officer');
INSERT INTO DESIGNATION (DESIG_CODE, DESIG_DESC) VALUES ('CLK', 'Clerk');
INSERT INTO DESIGNATION (DESIG_CODE, DESIG_DESC) VALUES ('HLP', 'Helper');
```

-- Inserting rows into the DEPARTMENT table

```
INSERT INTO DEPARTMENT (DEPT_CODE, DEPT_NAME) VALUES ('PER', 'Personnel');
INSERT INTO DEPARTMENT (DEPT_CODE, DEPT_NAME) VALUES ('PROD', 'Production');
INSERT INTO DEPARTMENT (DEPT_CODE, DEPT_NAME) VALUES ('PUR', 'Purchase');
INSERT INTO DEPARTMENT (DEPT_CODE, DEPT_NAME) VALUES ('FIN', 'Finance');
INSERT INTO DEPARTMENT (DEPT_CODE, DEPT_NAME) VALUES ('RES', 'Research');
```

-- Inserting rows into the EMPLOYEE table

```
INSERT INTO EMPLOYEE (EMP_CODE, EMP_NAME, DEPT_CODE, DESIG_CODE, SEX, ADDRESS, CITY,
STATE, PIN, BASIC, JN_DT)
```



```
VALUES ('E001', 'Arindam Chakraborty', 'FIN', 'MGR', 'M', '123 Ballygunge Place',  
'Kolkata', 'West Bengal', '700019', 60000, TO_DATE('2001-05-15', 'YYYY-MM-DD'));
```

```
INSERT INTO EMPLOYEE (EMP_CODE, EMP_NAME, DEPT_CODE, DESIG_CODE, SEX, ADDRESS, CITY,  
STATE, PIN, BASIC, JN_DT)
```

```
VALUES ('E002', 'Sourav Mukherjee', 'PROD', 'EXE', 'M', '56 Lake Town', 'Kolkata',  
'West Bengal', '700089', 40000, TO_DATE('2005-08-20', 'YYYY-MM-DD'));
```

```
INSERT INTO EMPLOYEE (EMP_CODE, EMP_NAME, DEPT_CODE, DESIG_CODE, SEX, ADDRESS, CITY,  
STATE, PIN, BASIC, JN_DT)
```

```
VALUES ('E003', 'Rina Sen', 'PUR', 'OFF', 'F', '78 Salt Lake', 'Kolkata', 'West  
Bengal', '700106', 35000, TO_DATE('2010-02-10', 'YYYY-MM-DD'));
```

```
INSERT INTO EMPLOYEE (EMP_CODE, EMP_NAME, DEPT_CODE, DESIG_CODE, SEX, ADDRESS, CITY,  
STATE, PIN, BASIC, JN_DT)
```

```
VALUES ('E004', 'Prasenjit Dutta', 'RES', 'CLK', 'M', '22 Park Street', 'Kolkata',  
'West Bengal', '700016', 25000, TO_DATE('2015-07-25', 'YYYY-MM-DD'));
```

```
INSERT INTO EMPLOYEE (EMP_CODE, EMP_NAME, DEPT_CODE, DESIG_CODE, SEX, ADDRESS, CITY,  
STATE, PIN, BASIC, JN_DT)
```

```
VALUES ('E005', 'Tanima Ghosh', 'PER', 'HLP', 'F', '45 Gariahat Road', 'Kolkata',  
'West Bengal', '700029', 20000, TO_DATE('2020-11-30', 'YYYY-MM-DD'));
```

```
INSERT INTO EMPLOYEE (EMP_CODE, EMP_NAME, DEPT_CODE, DESIG_CODE, SEX, ADDRESS, CITY,  
STATE, PIN, BASIC, JN_DT)
```

```
VALUES ('E006', 'Indrani Banerjee', 'RES', 'CLK', 'F', '10 Howrah Bridge Road',  
'Kolkata', 'West Bengal', '700001', 28000, TO_DATE('2000-03-10', 'YYYY-MM-DD'));
```

```
INSERT INTO EMPLOYEE (EMP_CODE, EMP_NAME, DEPT_CODE, DESIG_CODE, SEX, ADDRESS, CITY,  
STATE, PIN, BASIC, JN_DT)
```

```
VALUES ('E007', 'Rajib Dasgupta', 'PUR', 'EXE', 'M', '34 Dum Dum Road', 'Kolkata',  
'West Bengal', '700030', 45000, TO_DATE('2000-08-15', 'YYYY-MM-DD'));
```

```
INSERT INTO EMPLOYEE (EMP_CODE, EMP_NAME, DEPT_CODE, DESIG_CODE, SEX, ADDRESS, CITY,  
STATE, PIN, BASIC, JN_DT)
```

```
VALUES ('E008', 'Indrani Banerjee', 'RES', 'CLK', 'F', '10 Howrah Bridge Road',  
'Kolkata', 'West Bengal', '700001', 28000, TO_DATE('2000-03-10', 'YYYY-MM-DD'));
```

```
INSERT INTO EMPLOYEE (EMP_CODE, EMP_NAME, DEPT_CODE, DESIG_CODE, SEX, ADDRESS, CITY,  
STATE, PIN, BASIC, JN_DT)
```

```
VALUES ('E009', 'Rajib Dasgupta', 'PUR', 'EXE', 'M', '34 Dum Dum Road', 'Kolkata',  
'West Bengal', '700030', 45000, TO_DATE('2000-08-15', 'YYYY-MM-DD'));
```

```
INSERT INTO EMPLOYEE (EMP_CODE, EMP_NAME, DEPT_CODE, DESIG_CODE, SEX, ADDRESS, CITY, STATE, PIN, BASIC, JN_DT)
```

```
VALUES ('E010', 'Madhumita Sarkar', 'FIN', 'MGR', 'F', '15 Ballygunge Circular Road', 'Kolkata', 'West Bengal', '700019', 62000, TO_DATE('2012-11-23', 'YYYY-MM-DD'));
```

```
INSERT INTO EMPLOYEE (EMP_CODE, EMP_NAME, DEPT_CODE, DESIG_CODE, SEX, ADDRESS, CITY, STATE, PIN, BASIC, JN_DT)
```

```
VALUES ('E011', 'Anindita Chatterjee', 'PROD', 'MGR', 'F', '8 Southern Avenue', 'Kolkata', 'West Bengal', '700029', 65000, TO_DATE('2015-06-18', 'YYYY-MM-DD'));
```

```
INSERT INTO EMPLOYEE (EMP_CODE, EMP_NAME, DEPT_CODE, DESIG_CODE, SEX, ADDRESS, CITY, STATE, PIN, BASIC, JN_DT)
```

```
VALUES ('E012', 'Debarati Ghosh', 'RES', 'MGR', 'F', '5 College Street', 'Kolkata', 'West Bengal', '700012', 63000, TO_DATE('2018-04-05', 'YYYY-MM-DD'));
```

```
INSERT INTO EMPLOYEE (EMP_CODE, EMP_NAME, DEPT_CODE, DESIG_CODE, SEX, ADDRESS, CITY, STATE, PIN, BASIC, JN_DT)
```

```
VALUES ('E013', 'Ananya Mitra', 'PUR', 'OFF', 'F', '30 Park Circus', 'Kolkata', 'West Bengal', '700017', 35000, TO_DATE('2019-09-14', 'YYYY-MM-DD'));
```

```
INSERT INTO EMPLOYEE (EMP_CODE, EMP_NAME, DEPT_CODE, DESIG_CODE, SEX, ADDRESS, CITY, STATE, PIN, BASIC, JN_DT)
```

```
VALUES ('E014', 'Subhasish Roy', 'PER', 'CLK', 'M', '22 Elgin Road', 'Kolkata', 'West Bengal', '700020', 30000, TO_DATE('2022-01-28', 'YYYY-MM-DD'));
```

4)

-- Inserting rows without DEPT_CODE and BASIC

```
INSERT INTO EMPLOYEE (EMP_CODE, EMP_NAME, DESIG_CODE, SEX, ADDRESS, CITY, STATE, PIN, JN_DT)
```

```
VALUES ('E015', 'Rituparna Basu', 'EXE', 'F', '11 Gariahat Road', 'Kolkata', 'West Bengal', '700029', TO_DATE('2016-03-15', 'YYYY-MM-DD'));
```

```
INSERT INTO EMPLOYEE (EMP_CODE, EMP_NAME, DESIG_CODE, SEX, ADDRESS, CITY, STATE, PIN, JN_DT)
```

```
VALUES ('E016', 'Avik Bhattacharya', 'OFF', 'M', '42 Bhowanipore', 'Kolkata', 'West Bengal', '700025', TO_DATE('2017-07-12', 'YYYY-MM-DD'));
```

-- Inserting rows with BASIC set to 0

```
INSERT INTO EMPLOYEE (EMP_CODE, EMP_NAME, DEPT_CODE, DESIG_CODE, SEX, ADDRESS, CITY, STATE, PIN, BASIC, JN_DT)
```

```
VALUES ('E017', 'Paromita Sen', 'PROD', 'CLK', 'F', '9 Jadavpur', 'Kolkata', 'West Bengal', '700032', 0, TO_DATE('2013-05-20', 'YYYY-MM-DD'));
```

```
INSERT INTO EMPLOYEE (EMP_CODE, EMP_NAME, DEPT_CODE, DESIG_CODE, SEX, ADDRESS, CITY, STATE, PIN, BASIC, JN_DT)
```

```
VALUES ('E018', 'Anupam Ghoshal', 'FIN', 'HLP', 'M', '55 Salt Lake', 'Kolkata', 'West Bengal', '700091', 0, TO_DATE('2011-11-11', 'YYYY-MM-DD'));
```

5)

```
SELECT *
FROM EMPLOYEE
WHERE DEPT_CODE IS NULL;
```

EMP_CODE	EMP_NAME	DEPT_CODE	DESIG_CODE	SEX	ADDRESS	CITY	STATE	PIN	BASIC	JN_DT
E015	Rituparna Basu		EXE	F	11 Gariahat Road	Kolkata	West Bengal	700029		15-MAR-16
E016	Avik Bhattacharya		OFF	M	42 Bhowanipore	Kolkata	West Bengal	700025		12-JUL-17

6)

```
SELECT *
FROM EMPLOYEE
WHERE BASIC = 0;
```

EMP_CODE	EMP_NAME	DEPT_CODE	DESIG_CODE	SEX	ADDRESS	CITY	STATE	PIN	BASIC	JN_DT
E017	Paromita Sen	PROD	CLK	F	9 Jadavpur	Kolkata	West Bengal	700032	0	20-MAY-13
E018	Anupam Ghoshal	FIN	HLP	M	55 Salt Lake	Kolkata	West Bengal	700091	0	11-NOV-11

7)

```
SELECT *
FROM EMPLOYEE
WHERE BASIC IS NULL;
```

EMP_CODE	EMP_NAME	DEPT_CODE	DESIG_CODE	SEX	ADDRESS	CITY	STATE	PIN	BASIC	JN_DT
E015	Rituparna Basu		EXE	F	11 Gariahat Road	Kolkata	West Bengal	700029		15-MAR-16
E016	Avik Bhattacharya		OFF	M	42 Bhowanipore	Kolkata	West Bengal	700025		12-JUL-17

Difference Between Question 6 and 7 :

	NULL	'0'
Meaning	Indicates the absence of a value. It means that the data for that column is not available or has not been entered.	Indicates an explicitly assigned value of zero. This means that the value has been deliberately set to zero, often indicating a specific condition such as unpaid salary, voluntary work, or error in data entry.
Querying	SELECT * FROM EMPLOYEE WHERE BASIC IS NULL;	SELECT * FROM EMPLOYEE WHERE BASIC = 0;
Implications	Often requires careful handling in calculations and aggregations because operations on NULL result in NULL.	Can be included in calculations and treated as a numerical value.
Usage Context	Typically used when the value is genuinely unknown or not applicable.	Used when the value is specifically zero, indicating a known and intentional state.

8)

```
SELECT AVG(BASIC) AS Average_Basic
FROM EMPLOYEE;
```

AVERAGE_BASIC

36312.5

9)

```
UPDATE EMPLOYEE
SET BASIC = 0
WHERE BASIC IS NULL;
```

10)

AVERAGE_BASIC

32277.7777777777777777777777777777

The average value calculation depends on the data present in the column. When some values are set to 0, they contribute to the calculation of the average just like any other value. If there were NULL values before, they would have been excluded from the calculation, affecting the result.

For example, let's say we have three employees with the following basic salaries: 10000, 20000, and NULL. The average basic would be $(10000 + 20000) / 2 = 15000$.

If we replace the NULL value with 0 and recalculate the average, it would be $(10000 + 20000 + 0) / 3 = 10000$. This results in a different average because the NULL value is now considered as 0.

So, the difference in the average basic after replacing unassigned values with 0 is because those values now contribute to the calculation of the average.

11)

```
DELETE FROM EMPLOYEE
WHERE DEPT_CODE IS NULL;
```

12)

```
SELECT
    EMP_NAME,
    (BASIC + (BASIC * 0.5) + (BASIC * 0.4)) AS Net_Pay
FROM
    EMPLOYEE;
```

EMP_NAME	NET_PAY
Madhumita Sarkar	117800
Arindam Chakraborty	114000
Sourav Mukherjee	76000
Rina Sen	66500
Prasenjit Dutta	47500
Tanima Ghosh	38000
Indrani Banerjee	53200
Rajib Dasgupta	85500
Indrani Banerjee	53200
Rajib Dasgupta	85500
Anindita Chatterjee	123500
Debarati Ghosh	119700
Ananya Mitra	66500
Subhasish Roy	57000
Paromita Sen	0
Anupam Ghoshal	0

13)

```
SELECT UPPER(EMP_NAME) AS EMPLOYEE_NAME, BASIC
FROM EMPLOYEE
ORDER BY DEPT_CODE ASC;
```

EMPLOYEE_NAME	BASIC
ARINDAM CHAKRABORTY	60000
MADHUMITA SARKAR	62000
ANUPAM GHOSHAL	0
TANIMA GHOSH	20000
SUBHASISH ROY	30000
SOURAV MUKHERJEE	40000
PAROMITA SEN	0
ANINDITA CHATTERJEE	65000
RAJIB DASGUPTA	45000
RINA SEN	35000
ANANYA MITRA	35000
RAJIB DASGUPTA	45000
INDRANI BANERJEE	28000
DEBARATI GHOSH	63000
PRASENJIT DUTTA	25000
INDRANI BANERJEE	28000

14)

```
SELECT EMP_NAME, JN_DT
FROM EMPLOYEE
WHERE JN_DT > TO_DATE('2010-01-01', 'YYYY-MM-DD');
```

EMP_NAME	JN_DT
Madhumita Sarkar	23-NOV-12
Rina Sen	10-FEB-10
Prasenjit Dutta	25-JUL-15
Tanima Ghosh	30-NOV-20
Anindita Chatterjee	18-JUN-15
Debarati Ghosh	05-APR-18
Ananya Mitra	14-SEP-19
Subhasish Roy	28-JAN-22
Paromita Sen	20-MAY-13
Anupam Ghoshal	11-NOV-11

15)

```
SELECT COUNT(*) AS January_Join_Count
FROM EMPLOYEE
WHERE TO_CHAR(JN_DT, 'MM') = '01';
```

JANUARY_JOIN_COUNT

1

16)

```
SELECT
    MAX(BASIC) AS Max_Basic,
    MIN(BASIC) AS Min_Basic
FROM
    EMPLOYEE;
```

MAX_BASIC MIN_BASIC

65000 0

17)

```
SELECT COUNT(*) AS Female_Count
FROM EMPLOYEE
WHERE SEX = 'F';
```

FEMALE_COUNT

9

18)

```
UPDATE EMPLOYEE
SET CITY = UPPER(CITY);
```

19)

```
SELECT COUNT(DISTINCT CITY) AS Different_Cities_Count
FROM EMPLOYEE;
```

DIFFERENT_CITIES_COUNT

1

20)

```
SELECT EMP_CODE, EMP_NAME, DEPT_CODE, DESIG_CODE, SEX, ADDRESS, CITY, STATE, PIN,
BASIC, JN_DT
FROM EMPLOYEE
ORDER BY DEPT_CODE ASC, BASIC DESC;
```

EMP_CODE	EMP_NAME	DEPT_CODE	DESIG_CODE	SEX	ADDRESS	CITY	STATE	PIN	BASIC	JN_DT
E010	Madhumita Sarkar	FIN	MGR	F	15 Ballygunge Circular	KOLKATA	West Bengal	700019	62000	23-NOV-12
E001	Arindam Chakraborty	FIN	MGR	M	123 Ballygunge Place	KOLKATA	West Bengal	700019	60000	15-MAY-01
E018	Anupam Ghoshal	FIN	HLP	M	55 Salt Lake	KOLKATA	West Bengal	700091	0	11-NOV-11
E014	Subhasish Roy	PER	CLK	M	22 Elgin Road	KOLKATA	West Bengal	700020	30000	28-JAN-22
E005	Tanima Ghosh	PER	HLP	F	45 Gariahat Road	KOLKATA	West Bengal	700029	20000	30-NOV-20
E011	Anindita Chatterjee	PROD	MGR	F	8 Southern Avenue	KOLKATA	West Bengal	700029	65000	18-JUN-15
E002	Sourav Mukherjee	PROD	EXE	M	56 Lake Town	KOLKATA	West Bengal	700089	40000	20-AUG-05
E017	Paromita Sen	PROD	CLK	F	9 Jadavpur	KOLKATA	West Bengal	700032	0	20-MAY-13
E009	Rajib Dasgupta	PUR	EXE	M	34 Dum Dum Road	KOLKATA	West Bengal	700030	45000	15-AUG-00
E007	Rajib Dasgupta	PUR	EXE	M	34 Dum Dum Road	KOLKATA	West Bengal	700030	45000	15-AUG-00
E013	Ananya Mitra	PUR	OFF	F	30 Park Circus	KOLKATA	West Bengal	700017	35000	14-SEP-19
E003	Rina Sen	PUR	OFF	F	78 Salt Lake	KOLKATA	West Bengal	700106	35000	10-FEB-10
E012	Debarati Ghosh	RES	MGR	F	5 College Street	KOLKATA	West Bengal	700012	63000	05-APR-18
E008	Indrani Banerjee	RES	CLK	F	10 Howrah Bridge Road	KOLKATA	West Bengal	700001	28000	10-MAR-00
E006	Indrani Banerjee	RES	CLK	F	10 Howrah Bridge Road	KOLKATA	West Bengal	700001	28000	10-MAR-00
E004	Prasenjit Dutta	RES	CLK	M	22 Park Street	KOLKATA	West Bengal	700016	25000	25-JUL-15

Assignment 2

Shruti Mondal

Master of Computer Application

Roll No. : 002310503030

Semester : II

1)

```
SELECT
    DEPT_CODE,
    MIN(BASIC) AS Min_Basic,
    MAX(BASIC) AS Max_Basic,
    AVG(BASIC) AS Avg_Basic
FROM
    EMPLOYEE
GROUP BY
    DEPT_CODE;
```

DEPT_CODE	MIN_BASIC	MAX_BASIC	AVG_BASIC
PROD	0	65000	35000
FIN	0	62000	40666.666666667
PER	20000	30000	25000
PUR	35000	45000	40000
RES	25000	63000	36000

2)

```
SELECT DEPT_CODE, COUNT(*) AS female_count FROM EMPLOYEE WHERE SEX = 'F' GROUP BY
DEPT_CODE;
```

DEPT_CODE	FEMALE_COUNT
FIN	1
PER	1
PROD	2
PUR	2
RES	3

3)

```
SELECT DEPT_CODE,
    CITY,
    COUNT(*) AS num_employees
FROM EMPLOYEE
GROUP BY DEPT_CODE, CITY
ORDER BY DEPT_CODE, CITY;
```

DEPT_CODE	CITY	NUM_EMPLOYEES
FIN	KOLKATA	3
PER	KOLKATA	2
PROD	KOLKATA	3
PUR	KOLKATA	4
RES	KOLKATA	4

4)

```
SELECT DEPT_CODE,
    DESIG_CODE,
```

```

COUNT(*) AS num_employees
FROM EMPLOYEE
WHERE EXTRACT(YEAR FROM JN_DT) = 2000
GROUP BY DEPT_CODE, DESIG_CODE
ORDER BY num_employees ASC;

```

DEPT_CODE	DESIG_CODE	NUM_EMPLOYEES
RES	CLK	2
PUR	EXE	2

5)

```

SELECT DEPT_CODE,
       SUM(BASIC) AS total_basic
FROM EMPLOYEE
WHERE SEX = 'M'
GROUP BY DEPT_CODE
HAVING SUM(BASIC) > 50000
ORDER BY total_basic DESC;

```

DEPT_CODE	TOTAL_BASIC
PUR	90000
FIN	60000

6)

```

SELECT E.EMP_NAME,
       D.DESIG_DESC AS Designation,
       E.BASIC
FROM EMPLOYEE E
JOIN DESIGNATION D ON E.DESIG_CODE = D.DESIG_CODE;

```

EMP_NAME	DESIGNATION	BASIC
Madhumita Sarkar	Manager	62000
Arindam Chakraborty	Manager	60000
Sourav Mukherjee	Executive	40000
Rina Sen	Officer	35000
Prasenjit Dutta	Clerk	25000
Tanima Ghosh	Helper	20000
Indrani Banerjee	Clerk	28000
Rajib Dasgupta	Executive	45000
Indrani Banerjee	Clerk	28000
Rajib Dasgupta	Executive	45000
Anindita Chatterjee	Manager	65000
Debarati Ghosh	Manager	63000
Ananya Mitra	Officer	35000
Subhasish Roy	Clerk	30000
Paromita Sen	Clerk	0
Anupam Ghoshal	Helper	0

7)

```
SELECT E.EMP_NAME,  
       D.DESIG_DESC AS Designation,  
       DP.DEPT_NAME AS Department_Name,  
       E.BASIC  
FROM EMPLOYEE E  
JOIN DESIGNATION D ON E.DESIG_CODE = D.DESIG_CODE  
JOIN DEPARTMENT DP ON E.DEPT_CODE = DP.DEPT_CODE;
```

EMP_NAME	DESIGNATION	DEPARTMENT_NAME	BASIC
Tanima Ghosh	Helper	Personnel	20000
Subhasish Roy	Clerk	Personnel	30000
Sourav Mukherjee	Executive	Production	40000
Anindita Chatterjee	Manager	Production	65000
Paromita Sen	Clerk	Production	0
Rina Sen	Officer	Purchase	35000
Rajib Dasgupta	Executive	Purchase	45000
Rajib Dasgupta	Executive	Purchase	45000
Ananya Mitra	Officer	Purchase	35000
Arindam Chakraborty	Manager	Finance	60000
Madhumita Sarkar	Manager	Finance	62000
Anupam Ghoshal	Helper	Finance	0
Prasenjit Dutta	Clerk	Research	25000
Indrani Banerjee	Clerk	Research	28000
Indrani Banerjee	Clerk	Research	28000
Debarati Ghosh	Manager	Research	63000

8)

```
SELECT D.DEPT_CODE  
FROM DEPARTMENT D  
LEFT JOIN EMPLOYEE E ON D.DEPT_CODE = E.DEPT_CODE  
WHERE E.DEPT_CODE IS NULL;
```

9)

```
SELECT DISTINCT D.DEPT_NAME  
FROM DEPARTMENT D  
INNER JOIN EMPLOYEE E ON D.DEPT_CODE = E.DEPT_CODE;
```

DEPT_NAME

Finance

Purchase

Research

Personnel

Production

10)

```
SELECT D.DEPT_NAME  
FROM DEPARTMENT D  
JOIN EMPLOYEE E ON D.DEPT_CODE = E.DEPT_CODE  
GROUP BY D.DEPT_NAME  
HAVING COUNT(*) >= 10;
```

11)

```
SELECT EMP.DEPT_CODE
FROM EMPLOYEE EMP
WHERE EMP.BASIC = (
    SELECT MAX(BASIC)
    FROM EMPLOYEE
```

```
);
```

```
DEPT_CODE
```

```
-----
```

```
PROD
```

12)

```
SELECT DESIG.DESIG_DESC
FROM EMPLOYEE EMP
JOIN DESIGNATION DESIG ON EMP.DESIG_CODE = DESIG.DESIG_CODE
WHERE EMP.BASIC = (
    SELECT MAX(BASIC)
    FROM EMPLOYEE
```

```
);
```

```
DESIG_DESC
```

```
-----
```

```
Manager
```

13)

```
SELECT E.DEPT_CODE, COUNT(*) AS num_managers
FROM EMPLOYEE E
JOIN DESIGNATION D ON E.DESIG_CODE = D.DESIG_CODE
WHERE D.DESIG_DESC = 'Manager'
GROUP BY E.DEPT_CODE;
```

```
DEPT_CODE    NUM_MANAGERS
```

```
-----
```

```
PROD          1
```

```
FIN           2
```

```
RES           1
```

14)

```
SELECT BASIC
FROM EMPLOYEE
ORDER BY BASIC DESC
FETCH FIRST 1 ROW ONLY;
```

```
BASIC
```

```
-----
```

```
65000
```

15)

```
SELECT BASIC
FROM EMPLOYEE
ORDER BY BASIC
FETCH FIRST 1 ROW ONLY;
```

```
BASIC
```

```
-----
```

```
0
```

16)

```
SELECT DEPT_CODE
FROM EMPLOYEE
GROUP BY DEPT_CODE
ORDER BY SUM(BASIC) DESC
FETCH FIRST 1 ROW ONLY;
```

```
DEPT_CODE
-----
PUR
```

```
SELECT DEPT_CODE
FROM EMPLOYEE
GROUP BY DEPT_CODE
ORDER BY AVG(BASIC) DESC
FETCH FIRST 1 ROW ONLY;
```

```
DEPT_CODE
-----
FIN
```

```
SELECT DEPT_CODE
FROM EMPLOYEE
GROUP BY DEPT_CODE
ORDER BY COUNT(*) DESC
FETCH FIRST 1 ROW ONLY;
```

```
DEPT_CODE
-----
PUR
```

17)

```
INSERT INTO EMPLOYEE (EMP_CODE, EMP_NAME, DEPT_CODE, DESIG_CODE, SEX, ADDRESS, CITY,
STATE, PIN, BASIC, JN_DT)
VALUES ('EMP021', 'Sovan Sarkar', 'PUR', 'DES999', 'M', '123 Main Street', 'New
York', 'New York', '10001', 60000.00, TO_DATE('2023-10-01', 'YYYY-MM-DD'));
```

```
INSERT INTO EMPLOYEE (EMP_CODE, EMP_NAME, DEPT_CODE, DESIG_CODE, SEX, ADDRESS, CITY,
STATE, PIN, BASIC, JN_DT)
VALUES ('EMP022', 'Sathi Sardar', 'FIN', 'DES998', 'F', '456 Elm Street', 'Los
Angeles', 'California', '90001', 55000.00, TO_DATE('2023-10-15', 'YYYY-MM-DD'));
```

18)

```
DELETE FROM EMPLOYEE
WHERE DESIG_CODE NOT IN (SELECT DESIG_CODE FROM DESIGNATION);
```

19)

```
SELECT EMP_NAME
FROM EMPLOYEE E
WHERE SEX = 'F'
AND BASIC > (
    SELECT AVG(BASIC)
    FROM EMPLOYEE
    WHERE DEPT_CODE = E.DEPT_CODE
);
```

```
EMP_NAME
```

Madhumita Sarkar
Anindita Chatterjee
Debarati Ghosh

20)

```
SELECT COUNT(*) AS num_female_managers
FROM EMPLOYEE E
JOIN DESIGNATION D ON E.DESIG_CODE = D.DESIG_CODE
WHERE E.SEX = 'F' AND D.DESIG_DESC = 'Manager';
NUM_FEMALE_MANAGERS
```

Assignment 3

Shruti Mondal

Master of Computer Application

Roll No. : 002310503030

Semester : II

1)

```
CREATE TABLE departments (  
    department_code VARCHAR2(10) PRIMARY KEY,  
    department_name VARCHAR2(100) NOT NULL  
);  
  
CREATE TABLE employees (  
    employee_code NUMBER PRIMARY KEY,  
    name VARCHAR2(100) NOT NULL CHECK (name = UPPER(name)),  
    address VARCHAR2(255),  
    city VARCHAR2(100),  
    basic_salary NUMBER(8, 2) NOT NULL CHECK (basic_salary BETWEEN 5000 AND 9000),  
    join_date DATE DEFAULT SYSDATE,  
    grade CHAR(1) CHECK (grade IN ('A', 'B', 'C')),  
    department_code VARCHAR2(10),  
    CONSTRAINT fk_department  
        FOREIGN KEY (department_code)  
        REFERENCES departments(department_code)  
        ON DELETE SET NULL  
);  
  
CREATE TABLE leave_register (  
    leave_id NUMBER PRIMARY KEY,  
    employee_code NUMBER,  
    leave_type CHAR(2) CHECK (leave_type IN ('CL', 'EL', 'ML')),  
    from_date DATE NOT NULL,  
    to_date DATE NOT NULL,  
    CONSTRAINT fk_employee  
        FOREIGN KEY (employee_code)  
        REFERENCES employees(employee_code)  
        ON DELETE CASCADE  
);  
  
INSERT INTO departments (department_code, department_name) VALUES ('D001', 'Human  
Resources');  
INSERT INTO departments (department_code, department_name) VALUES ('D002',  
'Finance');  
INSERT INTO departments (department_code, department_name) VALUES ('D003', 'IT');  
INSERT INTO employees (employee_code, name, address, city, basic_salary, grade,  
department_code)  
VALUES (1, 'DIPENDU DHALI', 'Mahamaya Colony', 'Basirhat', 6000, 'A', 'D001');  
INSERT INTO employees (employee_code, name, address, city, basic_salary, grade,  
department_code)  
VALUES (2, 'SUBHAM MONDAL', 'Kamardanga', 'Basirhat', 7500, 'B', 'D002');  
INSERT INTO leave_register (leave_id, employee_code, leave_type, from_date, to_date)  
VALUES (1, 1, 'CL', TO_DATE('2023-01-01', 'YYYY-MM-DD'), TO_DATE('2023-01-05',  
'YYYY-MM-DD'));  
INSERT INTO leave_register (leave_id, employee_code, leave_type, from_date, to_date)
```

```
VALUES (2, 2, 'EL', TO_DATE('2023-02-01', 'YYYY-MM-DD'), TO_DATE('2023-02-10', 'YYYY-MM-DD'));
```

2)

```
INSERT INTO employees (employee_code, name, address, city, basic_salary, grade, department_code)
VALUES (3, 'ALICE BROWN', '789 Pine St', 'Los Angeles', 4000, 'A', 'D003');
```

ORA-02290: check constraint (SQL_IZFTSWLLOJVGWKKQCTEZHZFK.SYS_C00159655379) violated
ORA-06512: at "SYS.DBMS_SQL", line 1721

****Here we try to violating salary which have a range between 5000 to 9000**

3)

-- a) Create the view

```
CREATE VIEW dept_employees AS
SELECT employee_code, name, department_code AS dcode, basic_salary AS basic
FROM employees
WHERE department_code = 'D001';
```

-- b) Insert a row with a valid department code

```
INSERT INTO dept_employees (employee_code, name, dcode, basic)
VALUES (8, 'FRANK BLUE', 'D001', 6000);
```

-- Attempt to insert a row with an invalid department code (should fail)

```
INSERT INTO dept_employees (employee_code, name, dcode, basic)
VALUES (9, 'GRACE RED', 'D999', 6500);
```

ORA-02291: integrity constraint (SQL_IZFTSWLLOJVGWKKQCTEZHZFK.FK_DEPARTMENT) violated - parent key not found
ORA-06512: at "SYS.DBMS_SQL", line 1721

-- c) Find the newly inserted row in the original table

```
SELECT * FROM employees WHERE employee_code = 8;
```

EMPLOYEE_CODE	NAME	ADDRESS	CITY	BASIC_SALARY	JOIN_DATE	GRADE	DEPARTMENT_CODE
8	FRANK BLUE	-	-	6000	26-MAY-24	-	D001

-- d) Increment the basic salary by Rs. 100

```
UPDATE dept_employees
SET basic = basic + 100
WHERE employee_code = 8;
```

EMPLOYEE_CODE	NAME	ADDRESS	CITY	BASIC_SALARY	JOIN_DATE	GRADE	DEPARTMENT_CODE
8	FRANK BLUE	-	-	6100	26-MAY-24	-	D001

-- f) Delete the view

```
DROP VIEW dept_employees;
```

4)

--a)Creating a View

```
CREATE VIEW emp_leave_info AS
SELECT e.employee_code, e.name, d.department_name, e.basic_salary, l.leave_type,
l.from_date, l.to_date
FROM employees e
JOIN departments d ON e.department_code = d.department_code
LEFT JOIN leave_register l ON e.employee_code = l.employee_code;
```

--b) Inserting a Row into the View

```
INSERT INTO emp_leave_info (employee_code, name, department_name, basic_salary,
leave_type, from_date, to_date)
VALUES (10, 'HARRY GREEN', 'Human Resources', 7000, 'CL', TO_DATE('2024-05-28',
'YYYY-MM-DD'), TO_DATE('2024-05-30', 'YYYY-MM-DD'));
```

ORA-01779: cannot modify a column which maps to a non key-preserved table

--c) Trying to Increment 'basic_salary'

--d) Deleting the View

```
DROP VIEW emp_leave_info;
```

5)

-- a) Create a new table with specific records

```
CREATE TABLE dept_emp_7000 AS
SELECT e.employee_code, e.name, d.department_name, e.basic_salary
FROM employees e
JOIN departments d ON e.department_code = d.department_code
WHERE e.department_code = 'D001' AND e.basic_salary = 7000;
```

-- b) Insert employees with a basic salary greater than or equal to 7000

```
INSERT INTO dept_emp_7000 (employee_code, name, department_name, basic_salary)
SELECT e.employee_code, e.name, d.department_name, e.basic_salary
FROM employees e
JOIN departments d ON e.department_code = d.department_code
WHERE e.basic_salary >= 7000
AND NOT EXISTS (
    SELECT 1 FROM dept_emp_7000 de
    WHERE de.employee_code = e.employee_code
);
```

-- c) Alter the table to add a net_pay column

```
ALTER TABLE dept_emp_7000 ADD (net_pay NUMBER(10, 2));
```

-- d) Update net_pay with 1.5 * basic_salary

```
UPDATE dept_emp_7000  
SET net_pay = 1.5 * basic_salary;
```

-- e) Drop the net_pay column

```
ALTER TABLE dept_emp_7000 DROP COLUMN net_pay;
```

6)

-- Dropping the 'dept_emp_7000' table first

```
DROP TABLE dept_emp_7000;
```

-- Dropping the 'leave_register' table next

```
DROP TABLE leave_register;
```

-- Dropping the 'employees' table next

```
DROP TABLE employees;
```

-- Dropping the 'departments' table last

```
DROP TABLE departments;
```

Assignment 4

Shruti Mondal

Master of Computer Application

Roll No. : 002310503030

Semester : II

1)

--a) Create EMP table

```
CREATE TABLE EMP (  
    ECODE CHAR(10) PRIMARY KEY,  
    ENAME CHAR(50),  
    DCODE CHAR(10),  
    GRADE CHAR(1),  
    BASIC DECIMAL(10, 2),  
    JN_DT DATE  
);
```

--b) Insert some data

```
INSERT INTO EMP (ECODE, ENAME, DCODE, GRADE, BASIC, JN_DT) VALUES  
( 'E001', 'John Doe', 'D001', 'A', 50000.00, TO_DATE('2023-01-15', 'YYYY-MM-DD'));
```

```
INSERT INTO EMP (ECODE, ENAME, DCODE, GRADE, BASIC, JN_DT) VALUES  
( 'E002', 'Jane Smith', 'D002', 'B', 45000.00, TO_DATE('2023-02-20', 'YYYY-MM-DD'));
```

```
INSERT INTO EMP (ECODE, ENAME, DCODE, GRADE, BASIC, JN_DT) VALUES  
( 'E003', 'Jim Brown', 'D001', 'A', 55000.00, TO_DATE('2023-03-10', 'YYYY-MM-DD'));
```

```
INSERT INTO EMP (ECODE, ENAME, DCODE, GRADE, BASIC, JN_DT) VALUES  
( 'E004', 'Emily Davis', 'D003', 'C', 40000.00, TO_DATE('2023-04-05', 'YYYY-MM-DD'));
```

```
INSERT INTO EMP (ECODE, ENAME, DCODE, GRADE, BASIC, JN_DT) VALUES  
( 'E005', 'Michael Wilson', 'D004', 'B', 47000.00, TO_DATE('2023-05-12',  
'YYYY-MM-DD'));
```

2)

```
SELECT  
    ECODE AS "EMPLOYEE CODE",  
    ENAME AS "NAME",  
    DCODE AS "DEPT. CODE",  
    JN_DT AS "JOINING DATE"  
FROM  
    EMP;
```

EMPLOYEE CODE	NAME	DEPT. CODE	JOINING DATE
E001	John Doe	D001	2023-01-15
E002	Jane Smith	D002	2023-02-20
E003	Jim Brown	D001	2023-03-10
E004	Emily Davis	D003	2023-04-05
E005	Michael Wilson	D004	2023-05-12

3)

```
SELECT
    ECODE AS "EMPLOYEE CODE",
    ENAME AS "NAME",
    DCODE AS "DEPT. CODE",
    TO_CHAR(BASIC, '999,999,999') AS "BASIC",
    RPAD(GRADE, 10) AS "GRADE",
    TO_CHAR(JN_DT, 'MONTH DD, YYYY') AS "JOINING DATE"
FROM
    EMP;
```

EMPLOYEE CODE	NAME	DEPT. CODE	BASIC	GRADE	JOINING DATE
E001	John Doe	D001	50,000	A	JANUARY 15, 2023
E002	Jane Smith	D002	45,000	B	FEBRUARY 20, 2023
E003	Jim Brown	D001	55,000	A	MARCH 10, 2023
E004	Emily Davis	D003	40,000	C	APRIL 05, 2023
E005	Michael Wilson	D004	47,000	B	MAY 12, 2023

4)

--a) Show the display attributes of all the columns

```
DESCRIBE EMP;
```

Column	Null?	Type
ECODE	NOT NULL	CHAR(10)
ENAME		CHAR(50)
DCODE		CHAR(10)
GRADE		CHAR(1)
BASIC		NUMBER(10,2)
JN_DT		DATE

--b) Show the display attributes of a particular column

```
DESCRIBE EMP ECODE;
```

--c) Suppress the newly set attributes of 'JN-DT'

```
SELECT
    ECODE AS "EMPLOYEE CODE",
    ENAME AS "NAME",
    DCODE AS "DEPT. CODE",
    TO_CHAR(BASIC, '999,999,999') AS "BASIC",
    RPAD(GRADE, 10) AS "GRADE",
    JN_DT AS "JOINING DATE"
FROM
    EMP;
```

EMPLOYEE CODE	NAME	DEPT. CODE	BASIC	GRADE	JOINING DATE
E001	John Doe	D001	50,000	A	15-JAN-23
E002	Jane Smith	D002	45,000	B	20-FEB-23
E003	Jim Brown	D001	55,000	A	10-MAR-23
E004	Emily Davis	D003	40,000	C	05-APR-23
E005	Michael Wilson	D004	47,000	B	12-MAY-23

--d) Reset the newly set attributes of 'JN-DT'

```

SELECT
    ECODE AS "EMPLOYEE CODE",
    ENAME AS "NAME",
    DCODE AS "DEPT. CODE",
    TO_CHAR(BASIC, '999,999,999') AS "BASIC",
    RPAD(GRADE, 10) AS "GRADE",
    TO_CHAR(JN_DT, 'MONTH DD, YYYY') AS "JOINING DATE"

```

FROM

EMP;

EMPLOYEE CODE	NAME	DEPT. CODE	BASIC	GRADE	JOINING DATE
E001	John Doe	D001	50,000	A	JANUARY 15, 2023
E002	Jane Smith	D002	45,000	B	FEBRUARY 20, 2023
E003	Jim Brown	D001	55,000	A	MARCH 10, 2023
E004	Emily Davis	D003	40,000	C	APRIL 05, 2023
E005	Michael Wilson	D004	47,000	B	MAY 12, 2023

--e) Reset the newly set attributes of all columns

```

SELECT
    ECODE,
    ENAME,
    DCODE,
    BASIC,
    GRADE,
    TO_CHAR(JN_DT, 'MONTH DD, YYYY') AS JN_DT

```

FROM

EMP;

ECODE	ENAME	DCODE	BASIC	GRADE	JN_DT
E001	John Doe	D001	50000	A	JANUARY 15, 2023
E002	Jane Smith	D002	45000	B	FEBRUARY 20, 2023
E003	Jim Brown	D001	55000	A	MARCH 10, 2023
E004	Emily Davis	D003	40000	C	APRIL 05, 2023
E005	Michael Wilson	D004	47000	B	MAY 12, 2023

--f) Show the display attributes of all columns again
DESCRIBE EMP;

Column	Null?	Type
ECODE	NOT NULL	CHAR(10)
ENAME		CHAR(50)
DCODE		CHAR(10)
GRADE		CHAR(1)
BASIC		NUMBER(10,2)
JN_DT		DATE

5)

--a) Show the records from the EMP table in ascending order of DCODE, with DCODE value shown only for the first record of that department.

```
SELECT
    DCODE,
    ECODE AS "EMPLOYEE CODE",
    ENAME AS "NAME",
    GRADE AS "GRADE",
    TO_CHAR(BASIC, '999,999,999') AS "BASIC",
    TO_CHAR(JN_DT, 'MONTH DD, YYYY') AS "JOINING DATE"
FROM
    EMP
ORDER BY
    DCODE;
```

DCODE	EMPLOYEE CODE	NAME	GRADE	BASIC	JOINING DATE
D001	E001	John Doe	A	50,000	JANUARY 15, 2023
D001	E003	Jim Brown	A	55,000	MARCH 10, 2023
D002	E002	Jane Smith	B	45,000	FEBRUARY 20, 2023
D003	E004	Emily Davis	C	40,000	APRIL 05, 2023
D004	E005	Michael Wilson	B	47,000	MAY 12, 2023